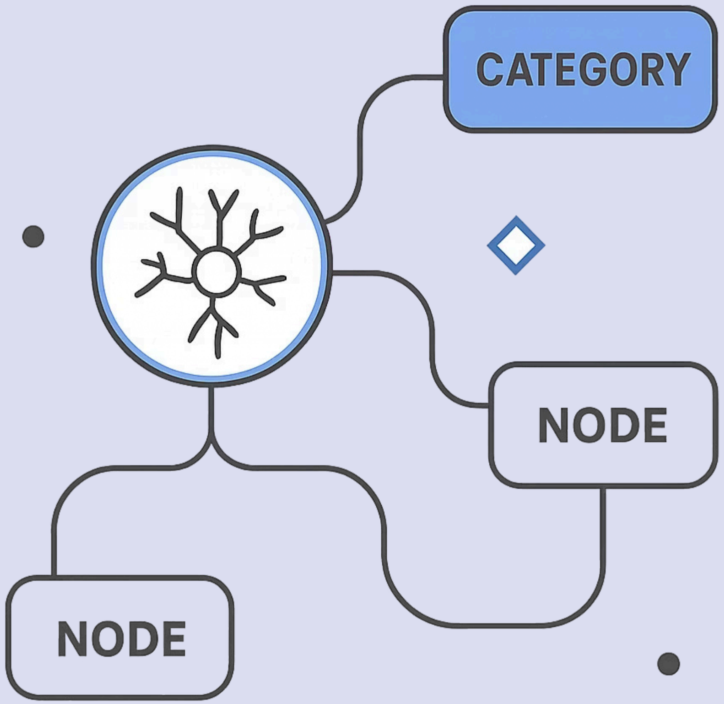
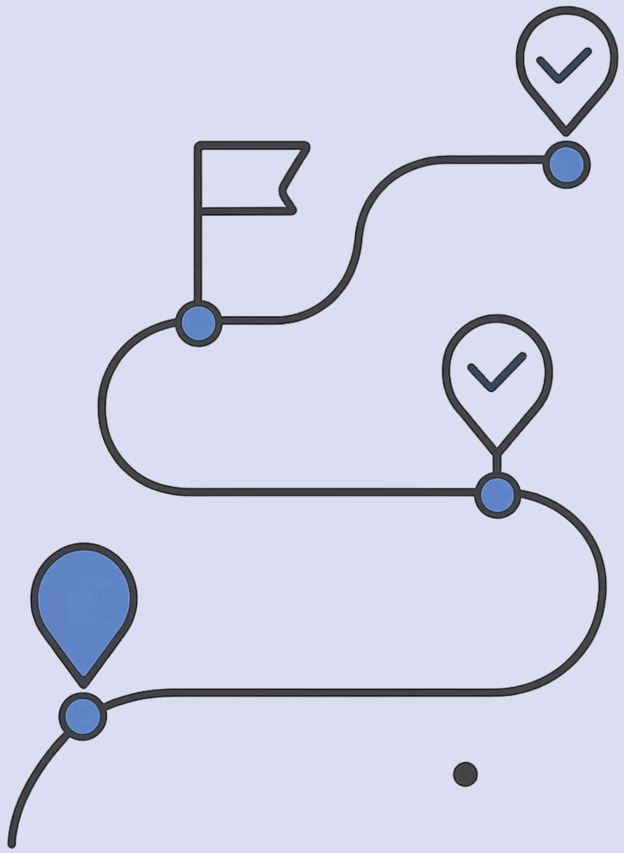
Classification Models Overview

From Basic Classifiers to Industry-Level Models



Classification Models Overview

Predicting categories instead of numbers



Where We Are

Regression Models

✅ Done — predicts numbers

Classification Models

🚀 Predicts categories

What is Classification?

Classification means predicting a class, label, or category.

Input	Prediction
Email text	Spam / Not Spam
Student performance	Pass / Fail
Medical data	Disease / No Disease
Image	Cat / Dog / Car

Regression vs Classification

Problem	Output	ML Type
Predict marks	Number	Regression
Predict pass/fail	Category	Classification
Predict salary	Number	Regression
Predict disease yes/no	Category	Classification
Predict loan approved/rejected	Category	Classification

Classification Output

Class Label

e.g. Spam

Probability

e.g. **0.87** probability of spam

Decision

Spam because probability > threshold

Many classifiers first estimate probability, then convert it into a class.

Classification Types

Binary

Two classes
e.g. Pass / Fail

Multiclass

More than two classes
e.g. Low / Medium / High

Multilabel

Multiple labels possible
e.g. Movie genres

Imbalanced

One class is rare
e.g. Fraud detection

Binary Classification

Binary classification has exactly two classes.

Problem	Classes
Email filtering	Spam / Not Spam
Student result	Pass / Fail
Loan approval	Approved / Rejected
Disease detection	Positive / Negative
Customer churn	Churn / Not Churn

Multiclass Classification

More than two classes — one input belongs to one class.

Problem	Classes
Grade prediction	A / B / C / D
Image classification	Cat / Dog / Horse
Customer segment	Low / Medium / High
Ticket priority	Low / Medium / Critical

Multilabel Classification

One input can have many labels simultaneously.

Movie	Labels
Movie A	Action, Thriller
Movie B	Comedy, Family
Movie C	Drama, Romance, Music

Unlike multiclass, a single record can belong to multiple classes at once.

Example Dataset

Student pass/fail prediction — target is categorical.

Study Hours	Attendance	Previous Score	Result
2	55	40	Fail
4	70	55	Pass
3	65	45	Fail
6	85	70	Pass
7	90	80	Pass

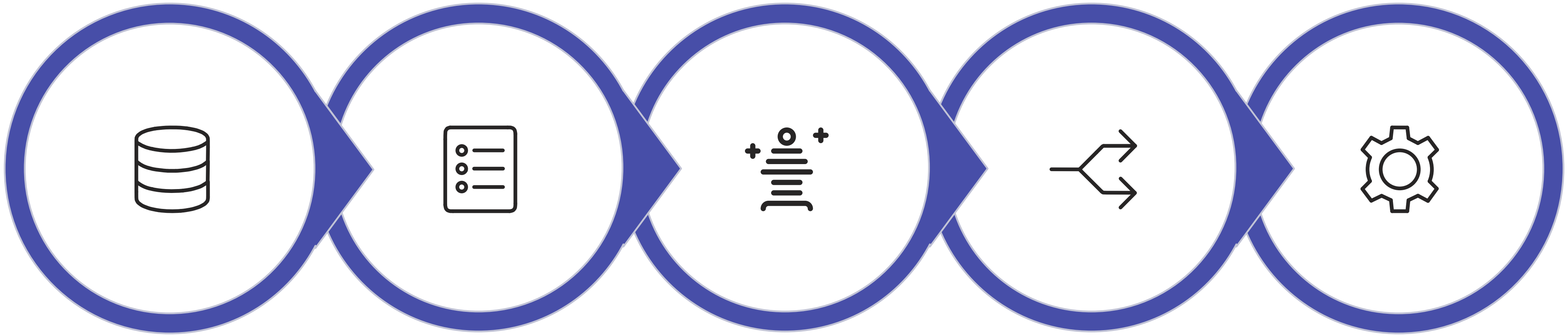
Feature and Target

Column	Role
Study Hours	Feature
Attendance	Feature
Previous Score	Feature
Result	Target

In classification, the target is a class label — not a number.

Example: Result = Pass or Fail

Classification Workflow



Collect Data

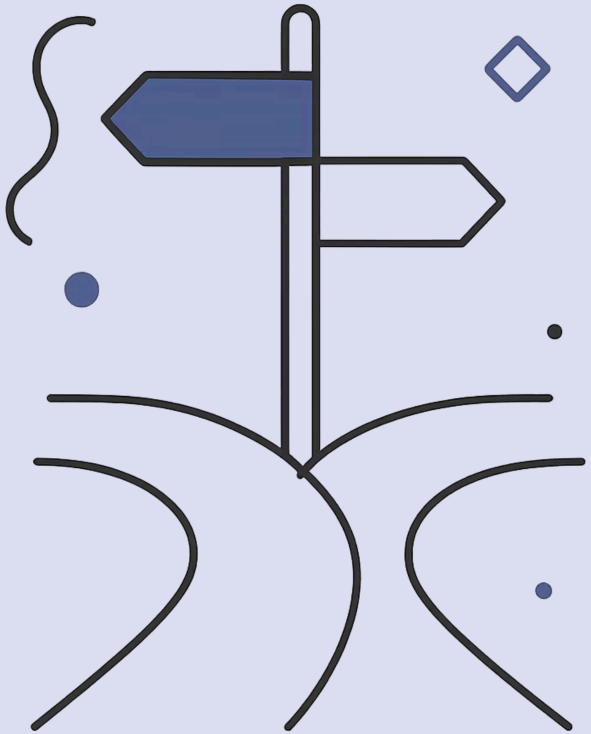
Choose Features

Encode Categorical

Train/Test Split

Train Model

Follow this workflow for every classification problem — from raw data to a refined, evaluated model.



PART 1

Core Classification Concepts

Before models, understand decisions and metrics.

Probability to Class

Many classifiers output a probability, then apply a decision rule.

Probability of Pass = 0.82

Threshold = 0.50

Decision Rule

If probability ≥ 0.50 → Pass

If probability < 0.50 → Fail

0.50 is called the threshold.

Threshold

Threshold decides class from probability. Changing threshold changes predictions.

Probability of Pass	Threshold	Prediction
0.82	0.50	Pass
0.63	0.50	Pass
0.41	0.50	Fail
0.20	0.50	Fail

Why Threshold Matters

In medical diagnosis, missing a disease case is dangerous. So we may use a lower threshold.

⚠ If disease probability ≥ **0.30** → Positive

This catches more true disease cases — but may increase false alarms.

Confusion Matrix

The foundation of classification evaluation for binary problems.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Confusion Matrix Example

Disease prediction results:

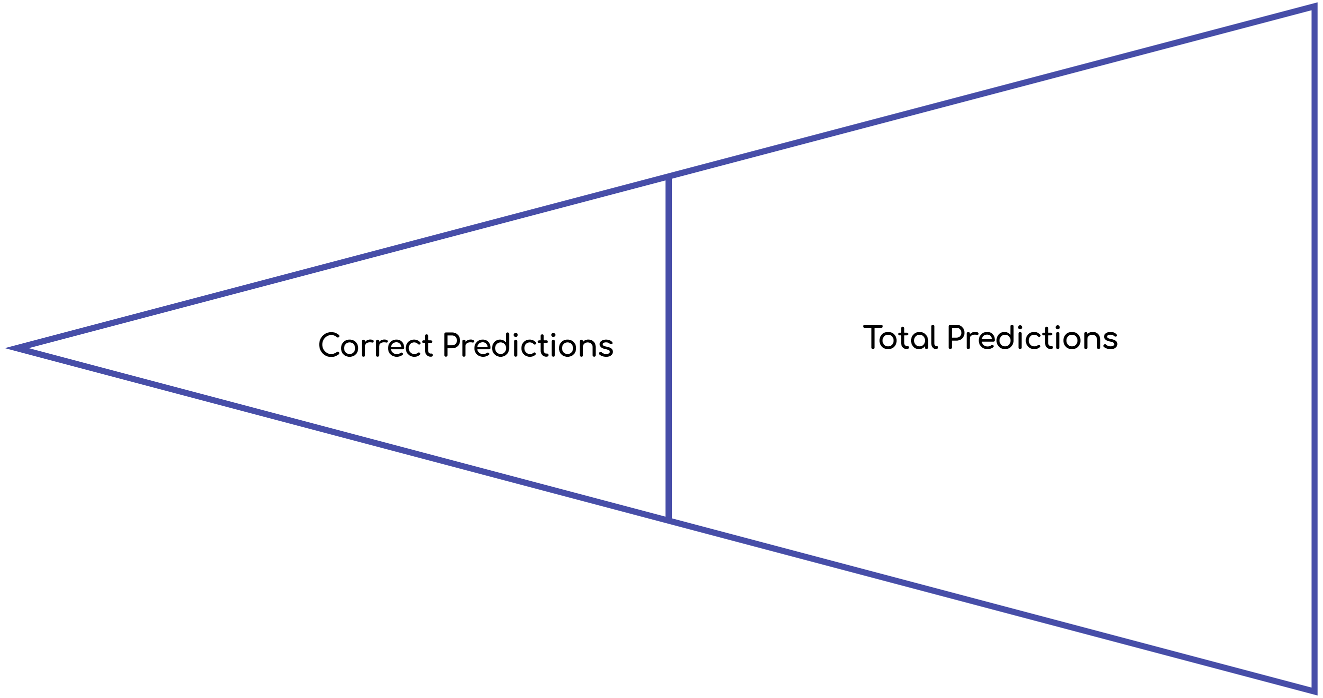
	Predicted Disease	Predicted No Disease
Actual Disease	80 (TP)	20 (FN)
Actual No Disease	10 (FP)	90 (TN)

Accuracy

How many predictions were correct overall?

Formula:
Accuracy = (TP + TN) / (TP + TN + FP + FN)

✅ Good when classes are balanced.



The Accuracy Problem

Scenario

Fraud cases are only **1%** of the dataset.

Model Predicts

Everyone is Not Fraud → Accuracy = 99%

Problem

Model catches zero fraud. Accuracy is misleading for imbalanced datasets.

Precision

Out of predicted positives, how many were actually positive?

Formula:
Precision = TP / (TP + FP)

Use when false positives are costly.

Example:
Spam detection should avoid marking important emails as spam.

Recall

Out of actual positives, how many did we catch?

Formula:
Recall = TP / (TP + FN)

Example:
Disease detection should not miss sick patients.

Use when false negatives are costly.

Precision vs Recall

Metric	Focus	Important When
Precision	Correctness of positive predictions	False positives are costly
Recall	Catching actual positives	False negatives are costly

Fraud Detection

Often needs high recall

Spam Filtering

Often needs good precision

F1 Score

F1 Score balances precision and recall into one combined metric.

Formula:
F1 = 2 × (Precision × Recall) / (Precision + Recall)

- Dataset is imbalanced
- Both precision and recall matter
- You need one combined score

ROC-AUC

Measures how well the model separates classes across thresholds. Higher AUC = better separation.

AUC = 0.5

Random guessing

AUC = 0.7

Fair

AUC = 0.8

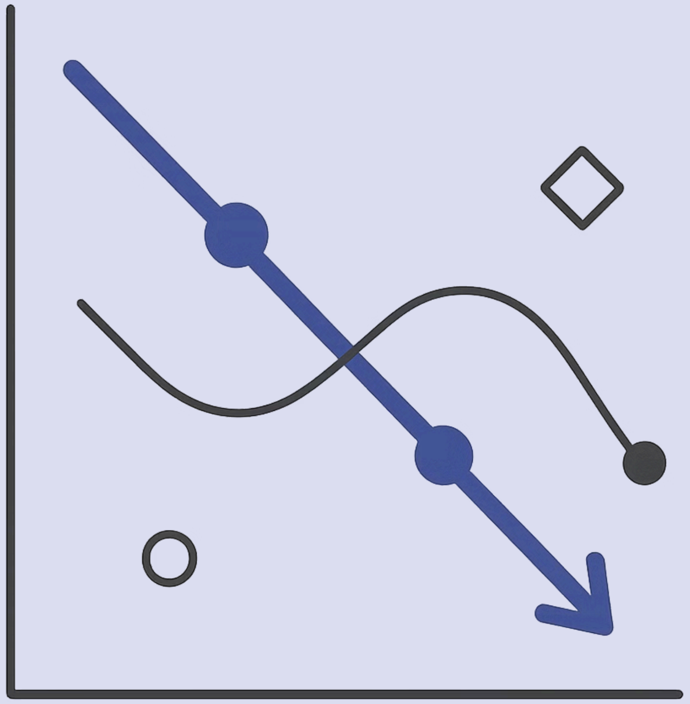
Good

AUC = 0.9+

Strong

Classification Metrics Summary

Metric	Best Used When
Accuracy	Classes are balanced
Precision	False positives are costly
Recall	False negatives are costly
F1 Score	Need balance between precision and recall
ROC-AUC	Need threshold-independent ranking quality



PART 2

Linear and Probability-Based Classifiers

Simple and interpretable classification models.

Model 1 — Logistic Regression

Despite the name, Logistic Regression is a classification model.

Problem	Classes
Pass prediction	Pass / Fail
Churn prediction	Churn / Not Churn
Disease prediction	Positive / Negative
Loan prediction	Approved / Rejected

Logistic Regression Idea

Linear Regression predicts any number. Logistic Regression predicts probability between 0 and 1.

Linear Score

$z = b_0 + b_1x_1 + b_2x_2 + \ldots$

Sigmoid Function

$p = 1 / (1 + e^{-z})$

Output

Probability = sigmoid(linear score)

Logistic Regression Decision

Probability of Pass

0.78

Threshold

0.50

Since $0.78 \geq 0.50$

Prediction: Pass

Logistic Regression Strengths

✅ Good For

- Binary classification
- Interpretable models
- Probability output
- Baseline classification model
- Linear decision boundaries

⚠️ Limitations

- Struggles with complex nonlinear patterns
- Needs feature engineering for curves/interactions
- Sensitive to outliers and scaling

Logistic Regression Use Case

Student pass/fail — output is Probability of Pass.

Study Hours

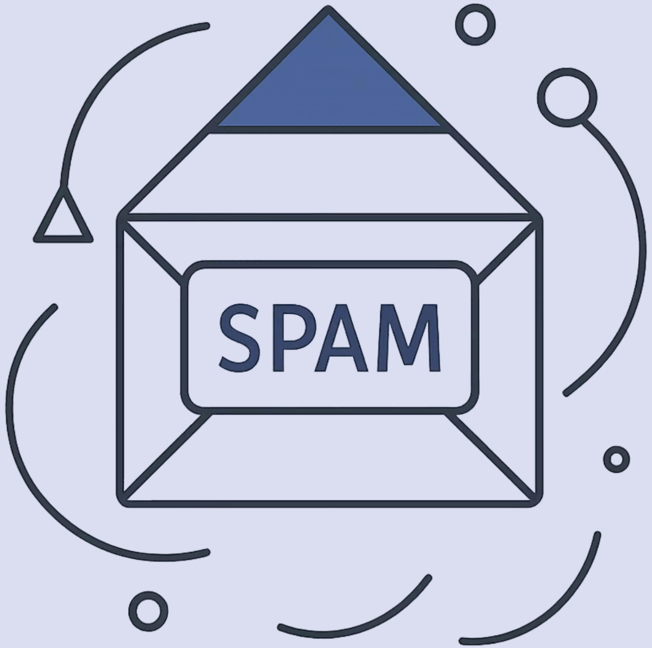
More hours may increase pass probability

Attendance

Higher attendance may increase pass probability

Previous Score

Strong previous score may increase pass probability



Model 2 — Naive Bayes

A probability-based classifier — fast and works well with text data.

Text Classification

Spam detection

Sentiment Analysis

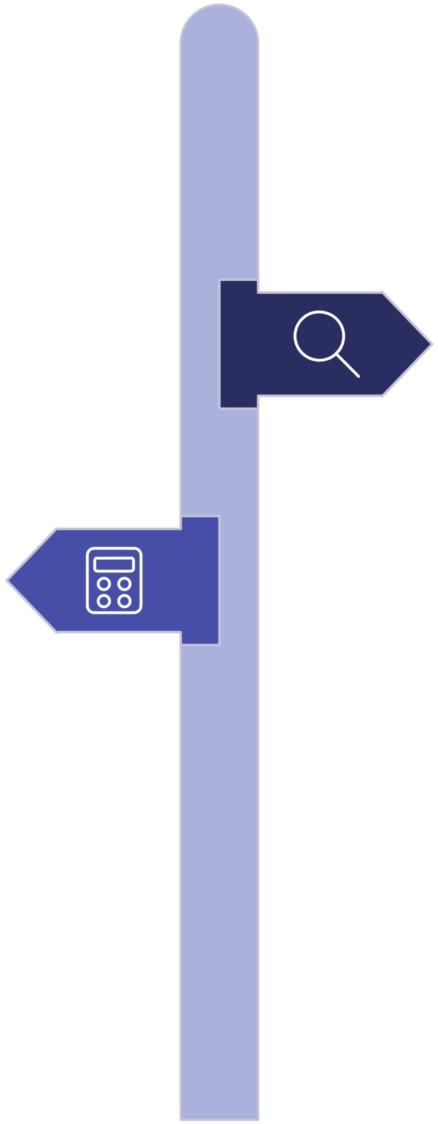
Positive / Negative

Document Classification

Topic prediction

Naive Bayes Idea

Given these features, which class is most likely?



Detect Keywords

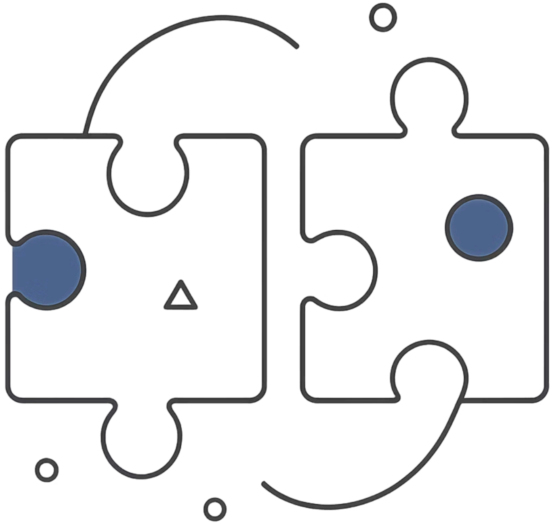
Find words like "free", "winner", "offer" in the email.

Estimate Spam

Combine feature probabilities to predict the Spam class.

Naive Bayes uses probability to assign the most likely class to each input.

Why "Naive"?



It assumes features are independent — each word contributes to the class independently of others.

⚠ This assumption is often not fully true — but the model can still work surprisingly well.

Naive Bayes Strengths

✅ Good For

- Text classification
- High-dimensional data
- Fast training
- Small datasets
- Baseline NLP tasks

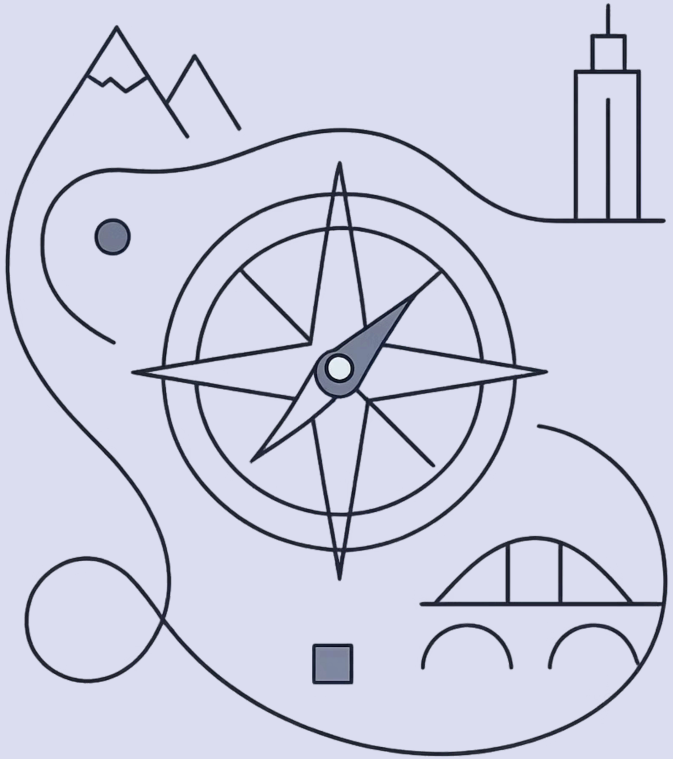
⚠️ Limitations

- Independence assumption is unrealistic
- Probability estimates may be rough
- Not ideal for complex feature interactions

PART 3

Distance-Based Classifier

Prediction using nearby examples.



Model 3 — KNN Classifier

K-Nearest Neighbors — a new record is classified based on nearby records.

Main Idea

Find the K nearest neighbours in the training data.

Decision

Assign the majority class among those neighbours.

Example

If nearest students mostly passed → predict Pass.

KNN Example

New student: Study Hours = 5, Attendance = 75

Neighbour	Result
Student A	Pass
Student B	Pass
Student C	Fail

Majority class: Pass

Prediction: Pass

Choosing K

K = number of neighbours. Choose using validation. Common values: K = 3, 5, 7.

Small K	Large K
Sensitive to noise	Smoother but may miss local patterns

KNN Strengths and Weaknesses

✅ Strengths

- Simple idea
- Can capture nonlinear boundaries
- No training complexity

⚠️ Weaknesses

- Needs feature scaling
- Slow for large datasets
- Sensitive to irrelevant features
- Choosing K matters